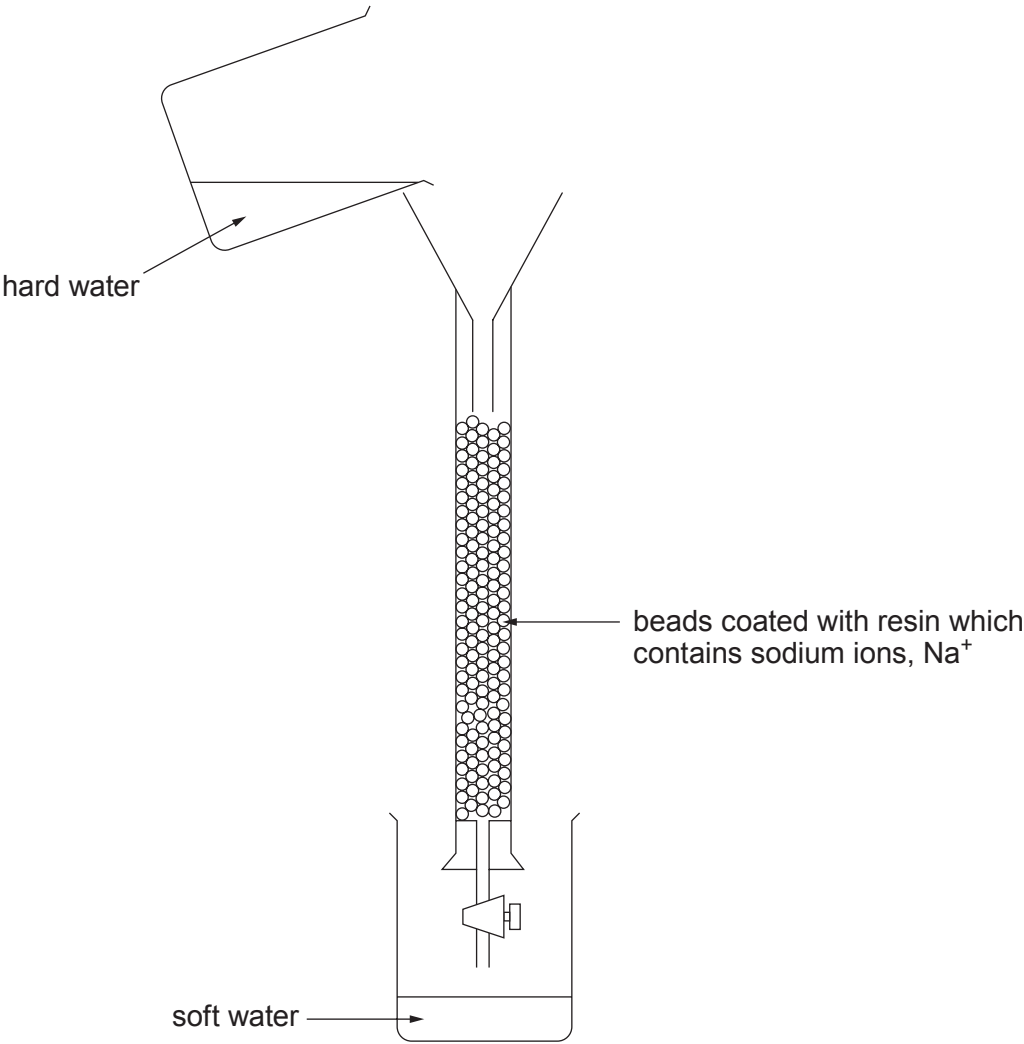


Examiner
only

7. (a) The diagram shows an ion exchange column used for softening permanent hard water.



(i) Explain how ion exchange works. [2]

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(ii) Explain why the ion exchange resin will stop working after continued use. Name a solution which can be passed through the resin to regenerate it. [2]

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(b) Describe and explain a method to test whether a water sample is soft water, temporary hard water or permanent hard water. [4]

(You do **not** need to include reference to fair testing in your answer)

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8

7. (a) Group 7 elements are known as the halogens. The following table shows the observations made when the first three members of the group react with hydrogen.

Halogen	Observations
fluorine	explodes in cold and dark
chlorine	explodes in sunlight
bromine	small explosion when ignited with a flame

- (i) Use your knowledge of electronic structure to explain why all the halogens react in a similar way and why they react more slowly on going down the group. [3]

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- (ii) Hydrogen fluoride is highly corrosive and can be used to etch glass which is mainly silicon dioxide.

Balance the symbol equation for the reaction between hydrogen fluoride and silicon dioxide. [1]



- (iii) Calcium fluoride reacts with sulfuric acid, H_2SO_4 , to produce calcium sulfate and hydrogen fluoride. Give the **symbol** equation for the reaction. [3]

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(b) Chlorine reacts with aluminium to produce aluminium chloride.

A sample of aluminium chloride of mass 26.70 g was found to contain 5.45 g of aluminium.
Calculate the simplest formula of this chloride of aluminium.

You **must** show your working.

[3]

$$A_r(\text{Al}) = 27$$

$$A_r(\text{Cl}) = 35.5$$

Formula

10



7. (a) Potassium has three stable isotopes – ^{39}K , ^{40}K and ^{41}K .

(i) Compare the nuclei of each of these isotopes. [1]

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(ii) Use the information to calculate the relative atomic mass, A_r , of potassium.

Record your answer to **three** significant figures. [3]

Isotope	Relative isotopic mass	% in sample
^{39}K	39	93.1
^{40}K	40	0.0122
^{41}K	41	6.88

$$A_r = \frac{(\text{mass} \times \% \text{ isotope 1}) + (\text{mass} \times \% \text{ isotope 2}) + (\text{mass} \times \% \text{ isotope 3})}{100}$$

$A_r =$



Examiner
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(b) Lithium lies above potassium in the Periodic Table.

(i) Give **two** similarities and **two** differences between the reactions of potassium and lithium with water. [2]

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(ii) Write a balanced symbol equation for the reaction between potassium and water. [3]

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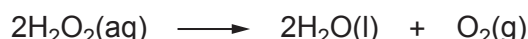
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7. A catalyst is a substance that causes significant changes to the rate of a chemical reaction. Catalysts interact with the reactants in a chemical reaction and eventually release the product allowing the catalyst to be recovered.

Enzymes are biological molecules which catalyse reactions in living organisms. Enzymes tend to have an optimum temperature of 37 °C.

Hydrogen peroxide is a colourless liquid with the chemical formula H_2O_2 . It has many uses including as a bleach and an antiseptic. It is also found in all animals and plants. The decomposition of hydrogen peroxide happens very slowly on its own.



The reaction can be catalysed by several substances including iron(III) oxide, Fe_2O_3 , manganese dioxide, MnO_2 , and the enzyme catalase.

Four catalysts were tested to determine which was the most efficient. An equal mass of each catalyst was added to equal volumes of hydrogen peroxide solution of equal concentration. The volume of oxygen produced in 1 minute was recorded. The experiment was repeated at different temperatures.

Temperature (°C)	Volume of oxygen produced in 1 minute (cm ³)			
	Catalyst W	Catalyst X	Catalyst Y	Catalyst Z
10	0.8	1.2	0.4	0.9
20	1.4	1.4	0.9	1.5
30	1.9	2.1	1.3	2.1
40	2.5	2.6	1.6	2.5
50	3.4	3.8	1.2	3.5
60	4.6	4.8	0.7	4.7
70	5.2	5.4	0.3	5.3



(a) Give the **number** of the statement which is **not** correct. [1]

- 1 Catalyst **W** is more efficient than **Y** but less efficient than **X** and **Z**
 - 2 Catalyst **X** is the most efficient of the four
 - 3 Catalyst **Y** is less efficient than all of the others
 - 4 Catalyst **Z** is more efficient than **Y** but less efficient than **W** and **X**
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(b) State which catalyst, **W**, **X**, **Y** or **Z**, could be catalase. Explain your answer. [3]

Catalyst

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(c) Three students made statements about the properties of catalysts.

- Katrina said that a catalyst does not get used up in the reaction but makes it happen faster.
- Jo said that the reactants are used up in less time because a catalyst makes a different product.
- Rhys said that adding a catalyst allows the reaction to happen with a lower activation energy, but the same products are made.

Which of the students' statements are correct? [1]

- A** Jo only
 - B** Rhys and Jo, but not Katrina
 - C** Katrina and Rhys, but not Jo
 - D** All three students
-



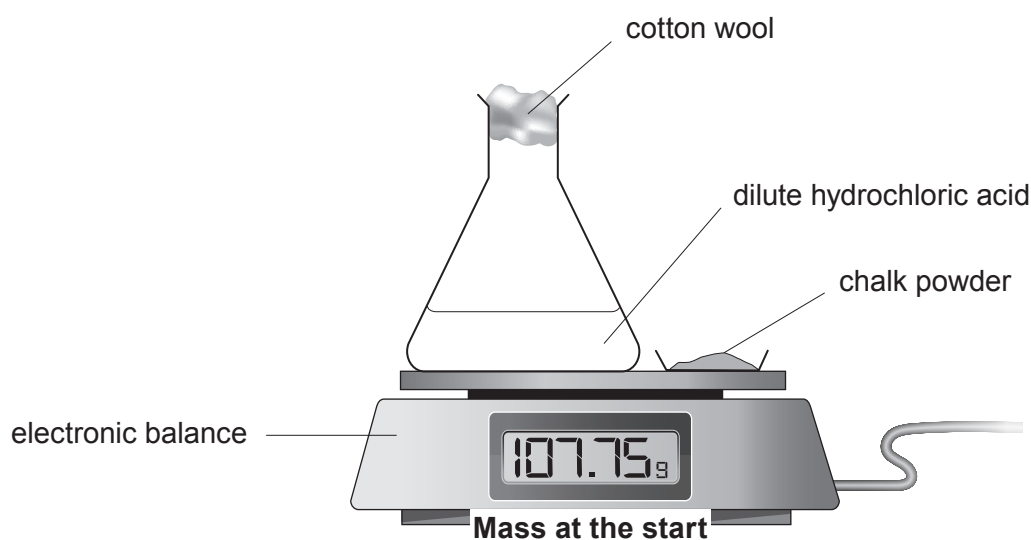
7. Chalk is a form of calcium carbonate, CaCO_3 .

A student carried out an experiment to investigate the rate of reaction between **powdered** chalk and **excess** dilute hydrochloric acid at 20°C .

Chalk reacts with dilute hydrochloric acid forming carbon dioxide gas.



The diagram shows the apparatus used.



- (a) State the purpose of the cotton wool.

[1]

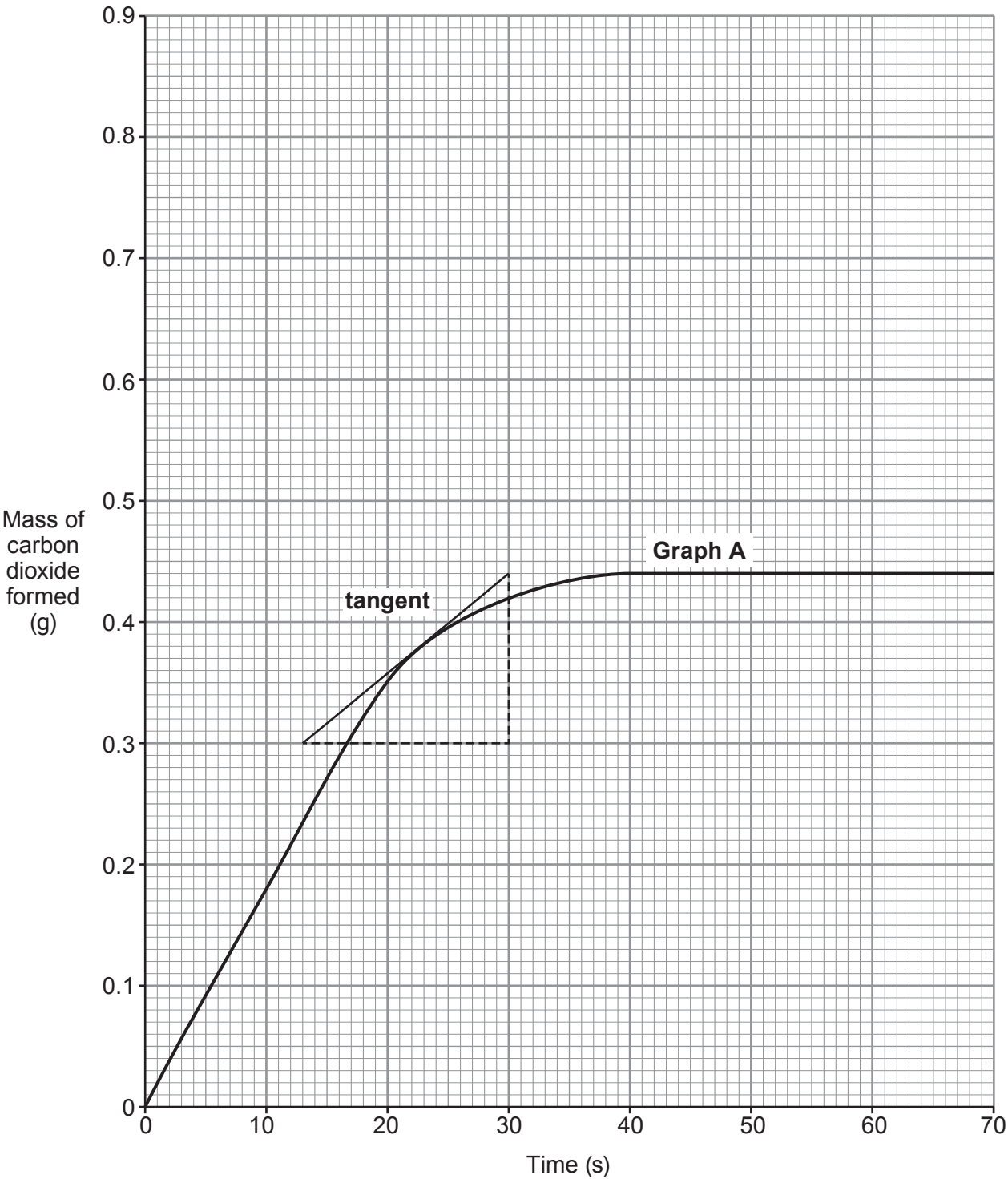
- (b) The student recorded the mass of the flask and contents every 10 seconds for 70 seconds.

Describe what the student must do to calculate the mass of carbon dioxide formed in the first 10 seconds.

[1]



- (c) Graph **A** shows the mass of carbon dioxide formed against time.
A tangent to the graph has been drawn at 23 s.



Examiner
only

Use the tangent and the equation below to calculate the rate of reaction at 23 s.
Show your working. Give your answer to **two** significant figures. [3]

rate = $\frac{\text{change in mass}}{\text{change in time}}$

Rate = g/s

- (d) The student repeated the experiment using the **same** mass of chalk powder as the original experiment but a **different concentration** of acid. The table shows the mass of carbon dioxide formed.

Time (s)	0	10	20	30	40	50	60	70
Mass of carbon dioxide formed (g)	0.00	0.11	0.20	0.28	0.35	0.40	0.43	0.44

- (i) Plot the results from the table on the grid opposite and draw a suitable line.
Label this graph **B**. [3]

- (ii) Use particle theory to explain why the rate of this reaction is lower than that in the original experiment. [3]

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- (e) Sketch on the grid the curve you would expect if the experiment was repeated using the **original** acid and **twice** the mass of chalk. Assume that the acid is still in excess.
Label this graph **C**. [1]

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